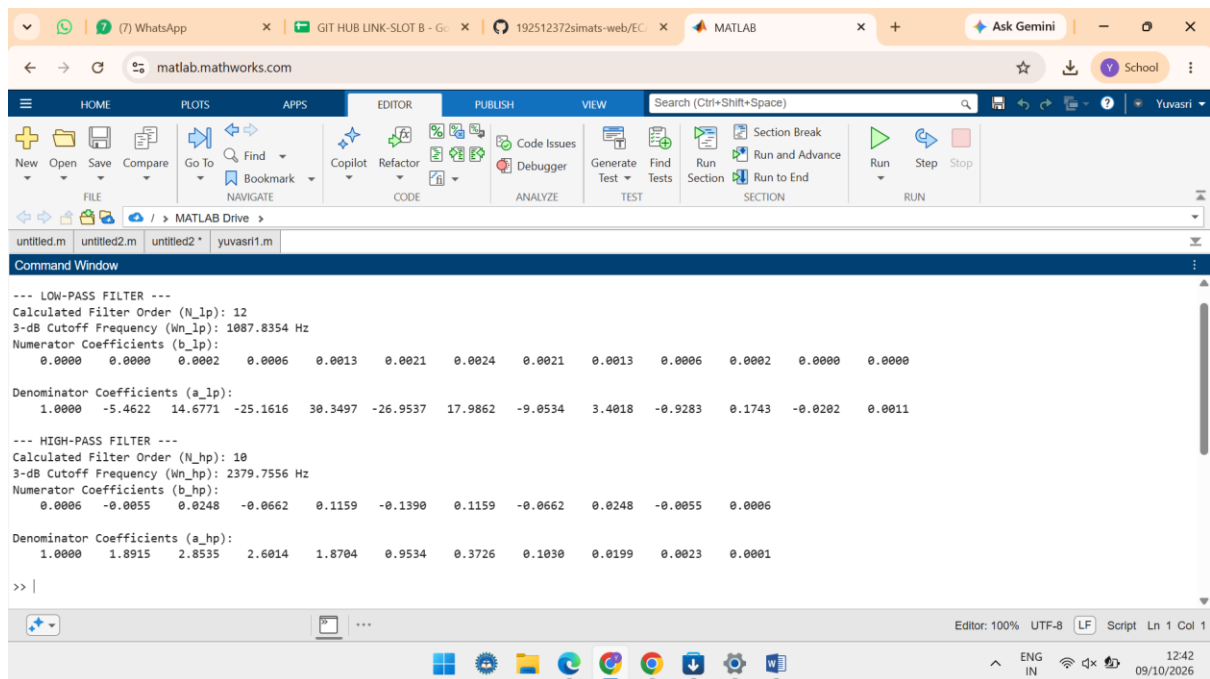
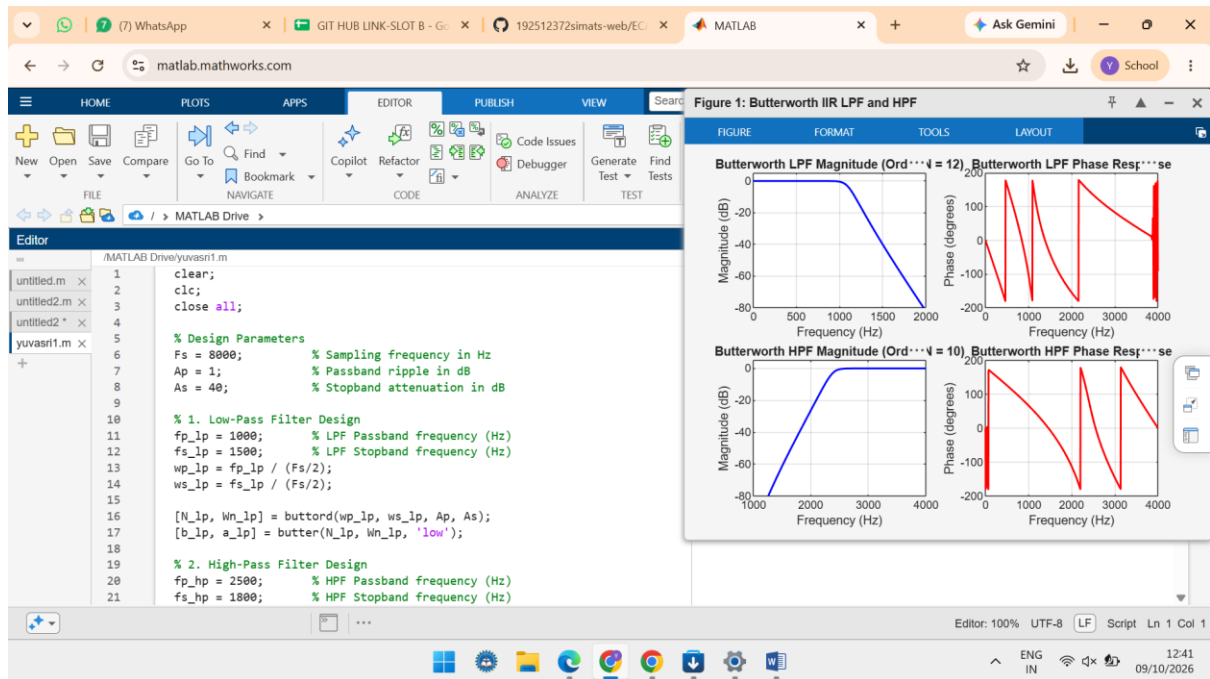


1.Design, Realization and Performance Analysis of Butterworth IIR Low-Pass and High-Pass Digital Filters Using MATLAB



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Save As Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

=====
FILTER ORDER COMPARISON
=====
Butterworth Order      = 12
Chebyshev Type-I Order = 6
Elliptic Order         = 4
=====

BUTTERWORTH COEFFICIENTS
=====
Numerator b =
1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator a =
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053
=====

```

12:56 09/10/2026

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Save As Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

=====
CHEBYSHEV TYPE-I COEFFICIENTS
=====
Numerator b =
1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

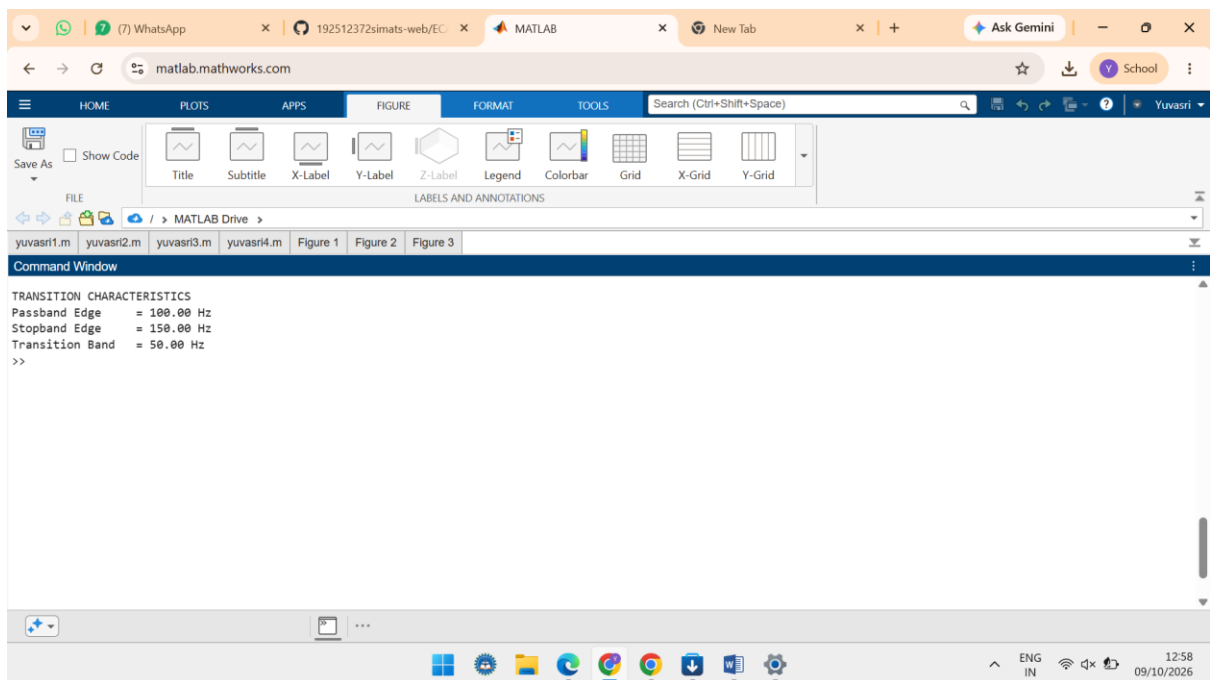
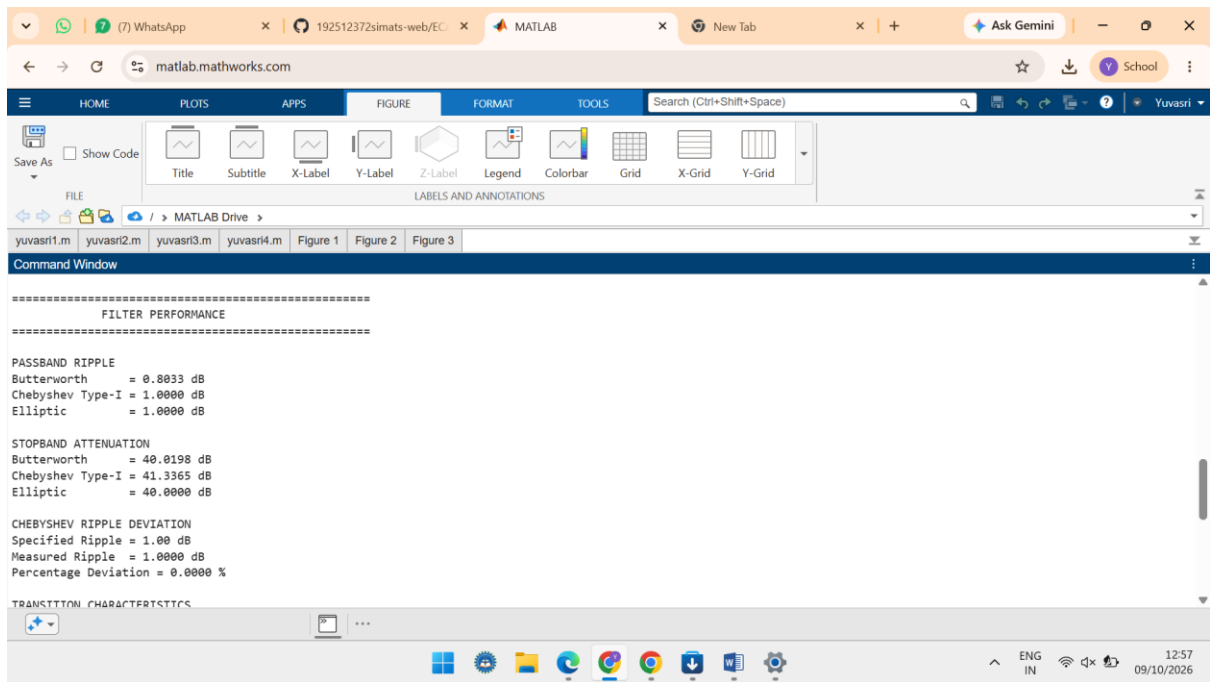
Denominator a =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586
=====

ELLIPTIC COEFFICIENTS
=====
Numerator b =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator a =
    1.0000   -3.0330    3.8118   -2.2911    0.5554
=====

```

12:56 09/10/2026



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Save As Show Code

FILE

Labels and Annotations: Title, Subtitle, X-Label, Y-Label, Z-Label, Legend, Colorbar, Grid, X-Grid, Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```
=====
FILTER ORDER COMPARISON
=====
Butterworth Order      = 12
Chebyshev Type-I Order = 6
Elliptic Order         = 4
=====

BUTTERWORTH COEFFICIENTS
=====
Numerator b =
1.0e-03 *
    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator a =
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053
=====

CHEBYSHEV TYPE-I COEFFICIENTS
=====
```

Windows Taskbar: ENG IN, 12:54, 09/10/2026